FINAL PROJECT

PROJECT TASKS

- GEM5 + NVMAIN BUILD-UP (40%) → in the tutorial below
- Enable L3 last level cache in GEM5 + NVMAIN (15%)
- Config last level cache to 2-way and full-way associative cache and test performance (15%)
  - You need to run the benchmark Quicksort in 2-way and full-way.
- Modify last level cache policy based on frequency based replacement policy (15%)
- Test the performance of write back and write through policy based on 4-way associative cache with isscc_pcm(15%)
  - You need to run the benchmark multiply in write through and write back.
- Bonus (10%)
  - Design last level cache policy to reduce the energy consumption of pcm_based main memory
  - Baseline:LRU

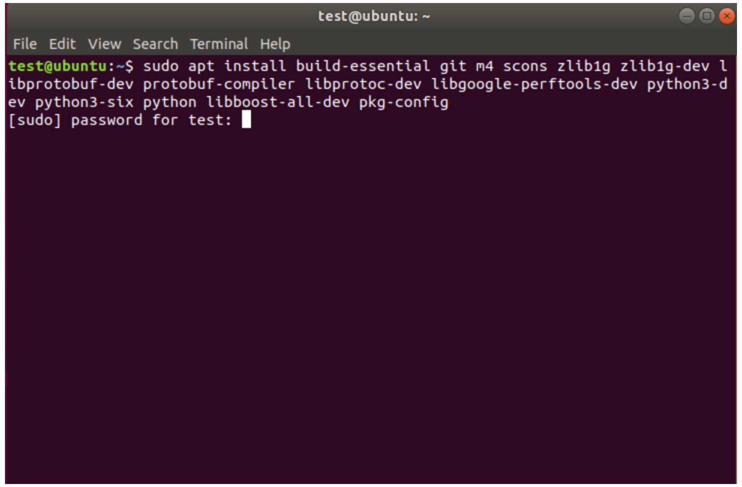
NVMAIN+GEM5 TUTORIAL

VIRTUAL MACHINE

- Please use Ubuntu 18.04
- Others settings
  - Memory space:4GB
  - Disk space :20GB

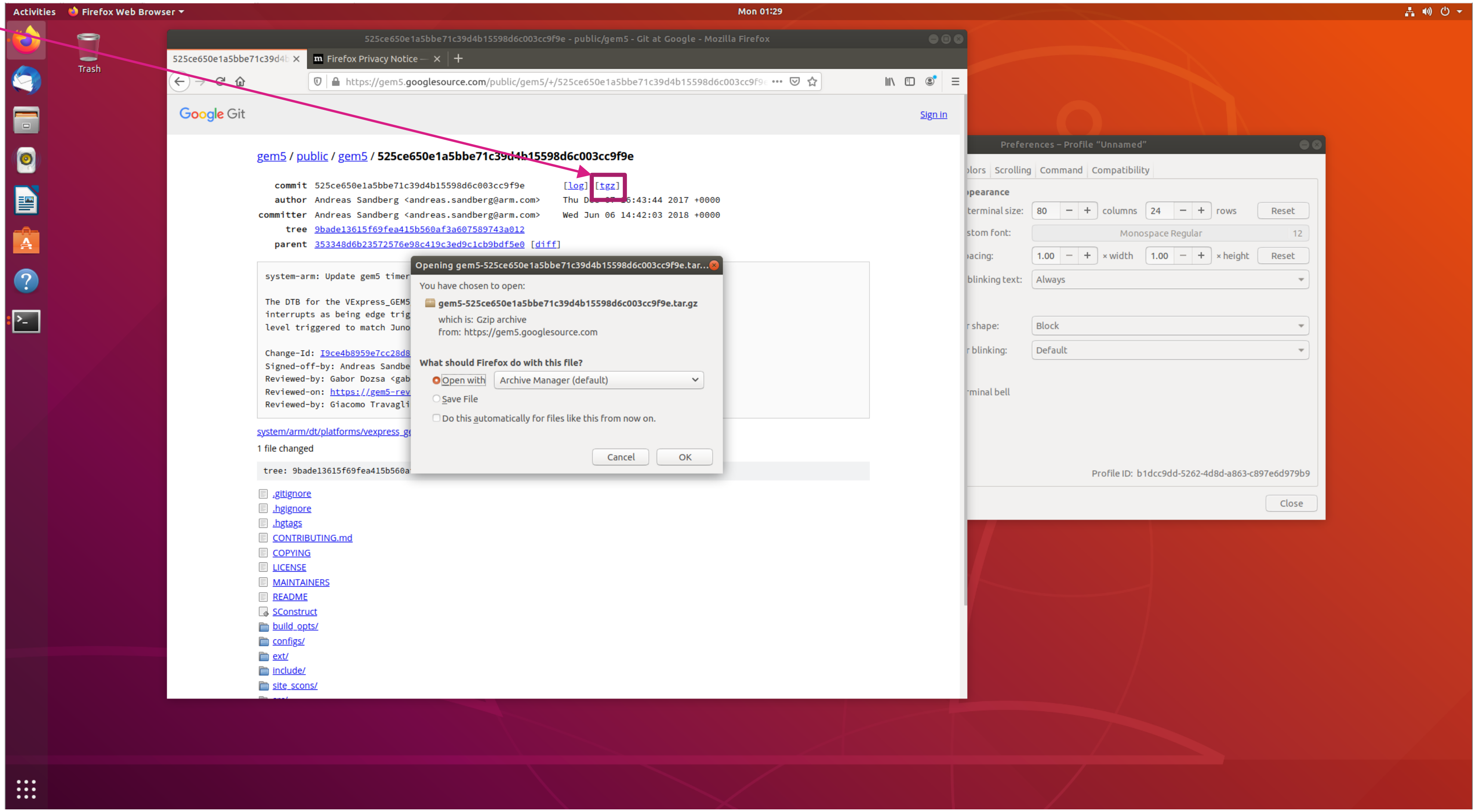
INSTALL BUILD TOOLS

- sudo apt install build-essential git m4 scons zlib1g zlib1g-dev libprotobuf-dev protobuf-compiler libprotoc-dev libgoogle-perftools-dev python3-dev python3-six python libboost-all-dev pkg-config

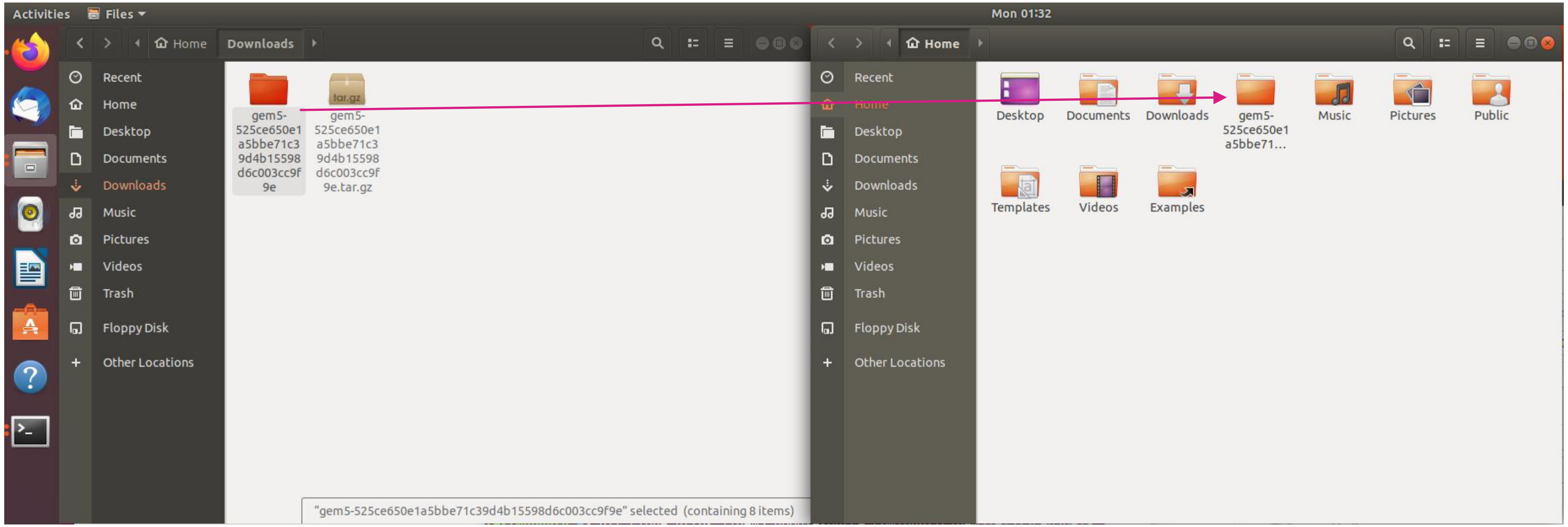


INSTALL GEM5

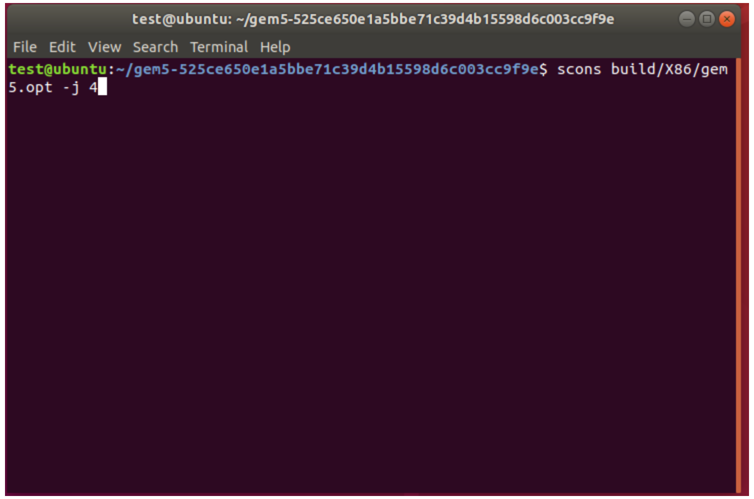
- https://gem5.googlesource.com/public/gem5/+/525ce650e1a5bbe71c39d4b15598d6c003cc9f9e
- Press tgz Download

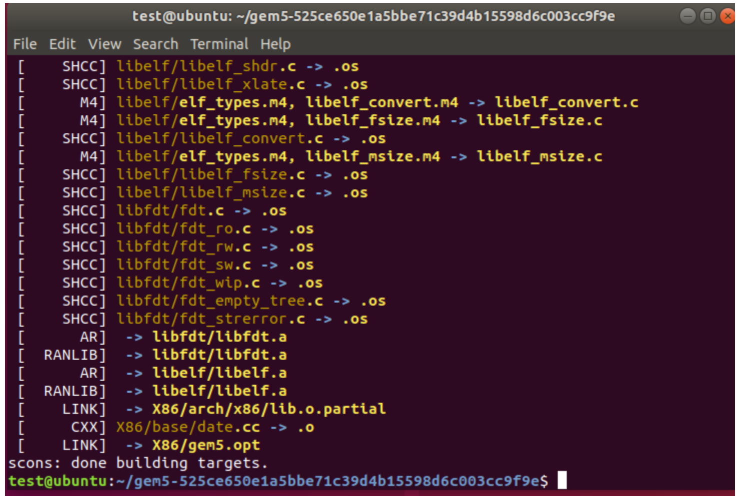


MOVING IT FROM DOWNLOADS TO /HOME



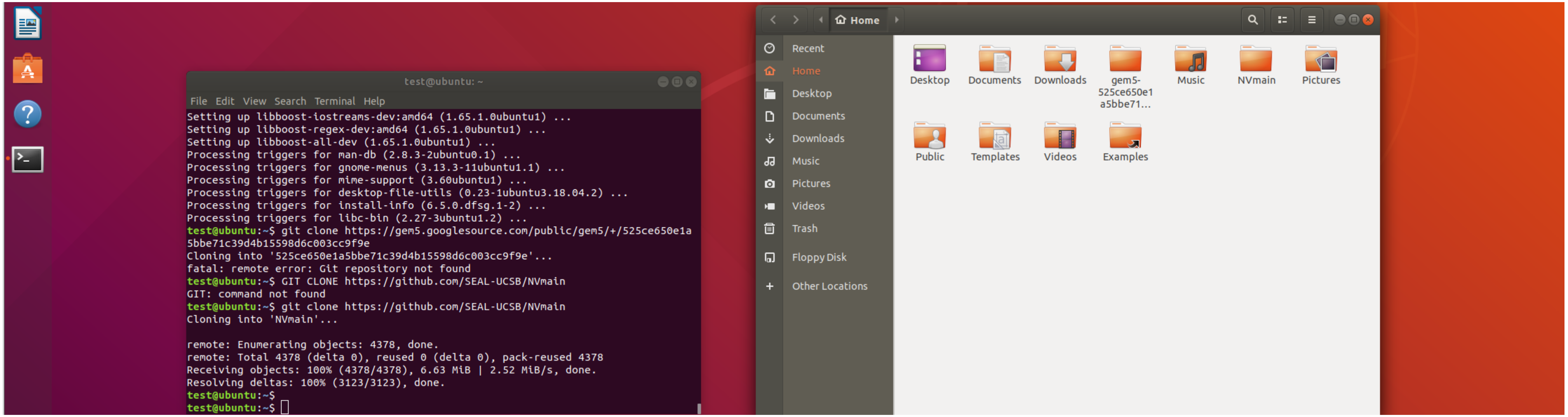
COMPILE GEM5





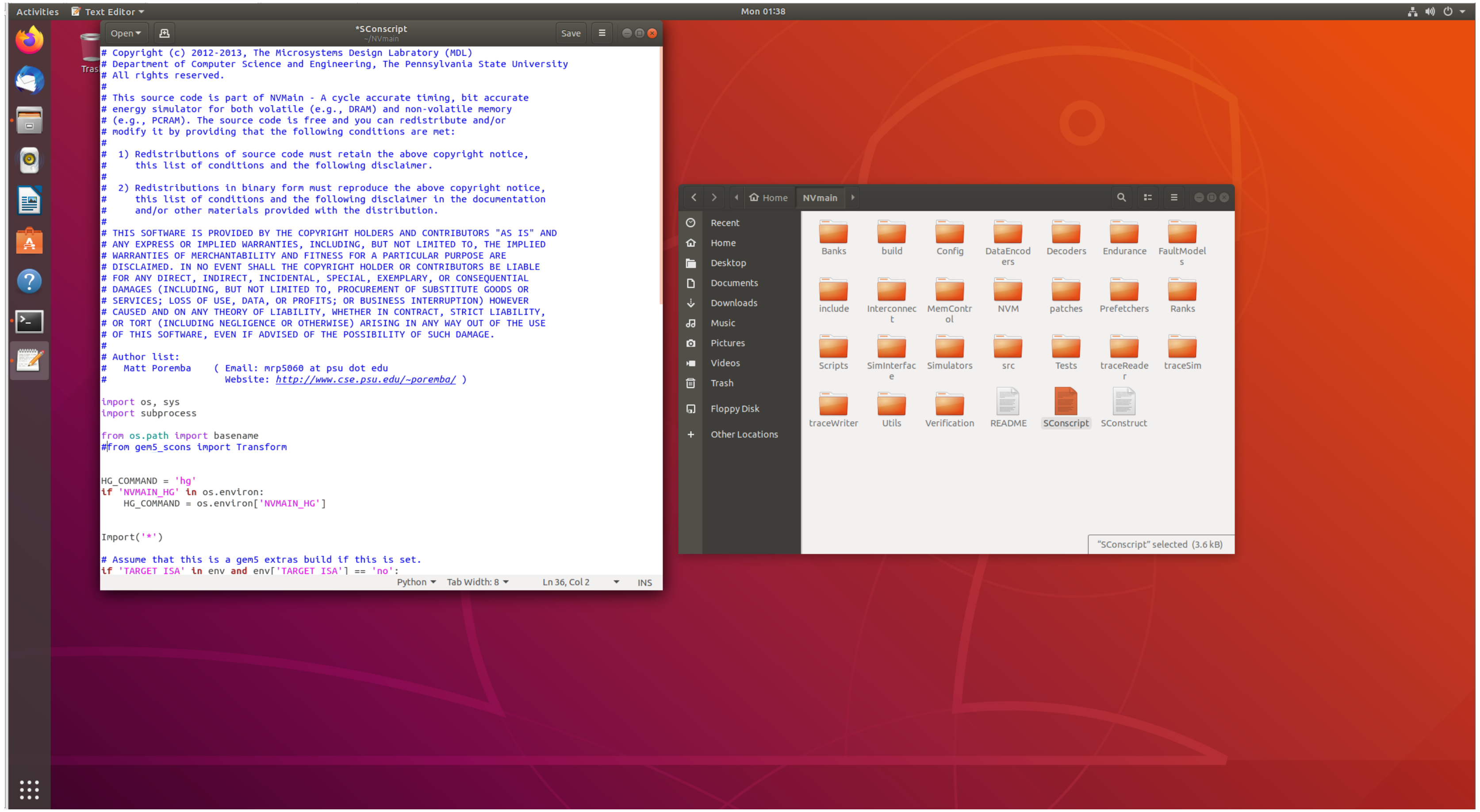
GIT INSTALL NVMAIN

- git clone https://github.com/SEAL-UCSB/NVmain
- Nvmain and gem5 should be in the /Home together.



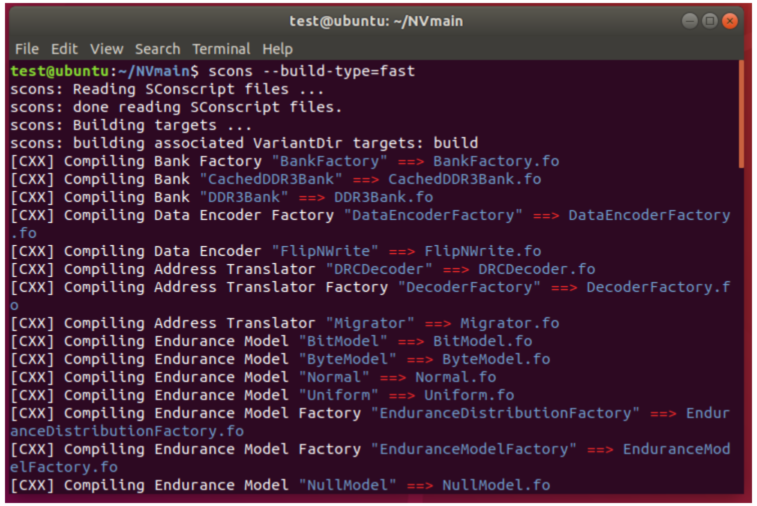
MODIFY SCONSCRIPT

- Enter the nvmain folder, click on SConscript, and annotate the 36 lines of from gem5_scons import Transform (remember to archive)



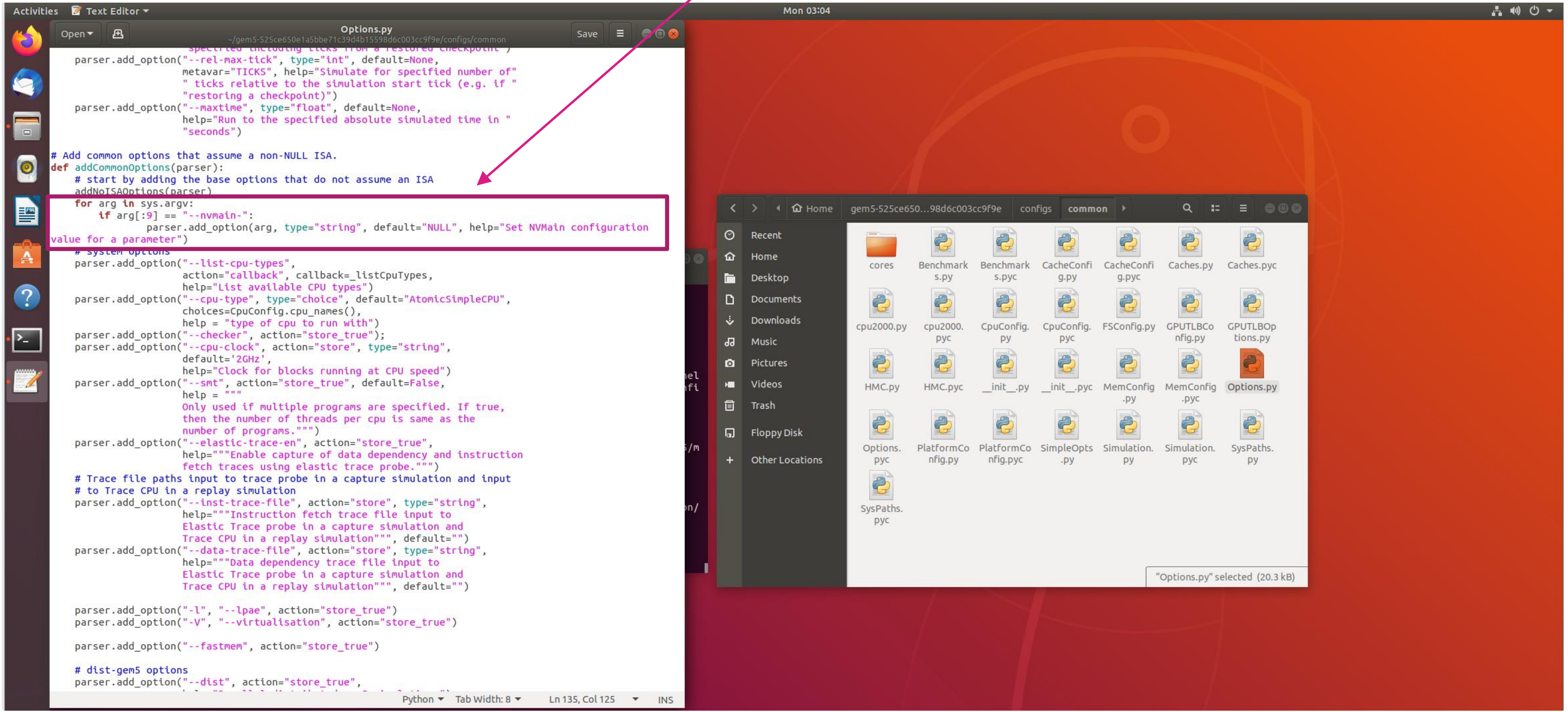
COMPILE NVMAIN

- In ./Nvmain Type: scons --build-type=fast



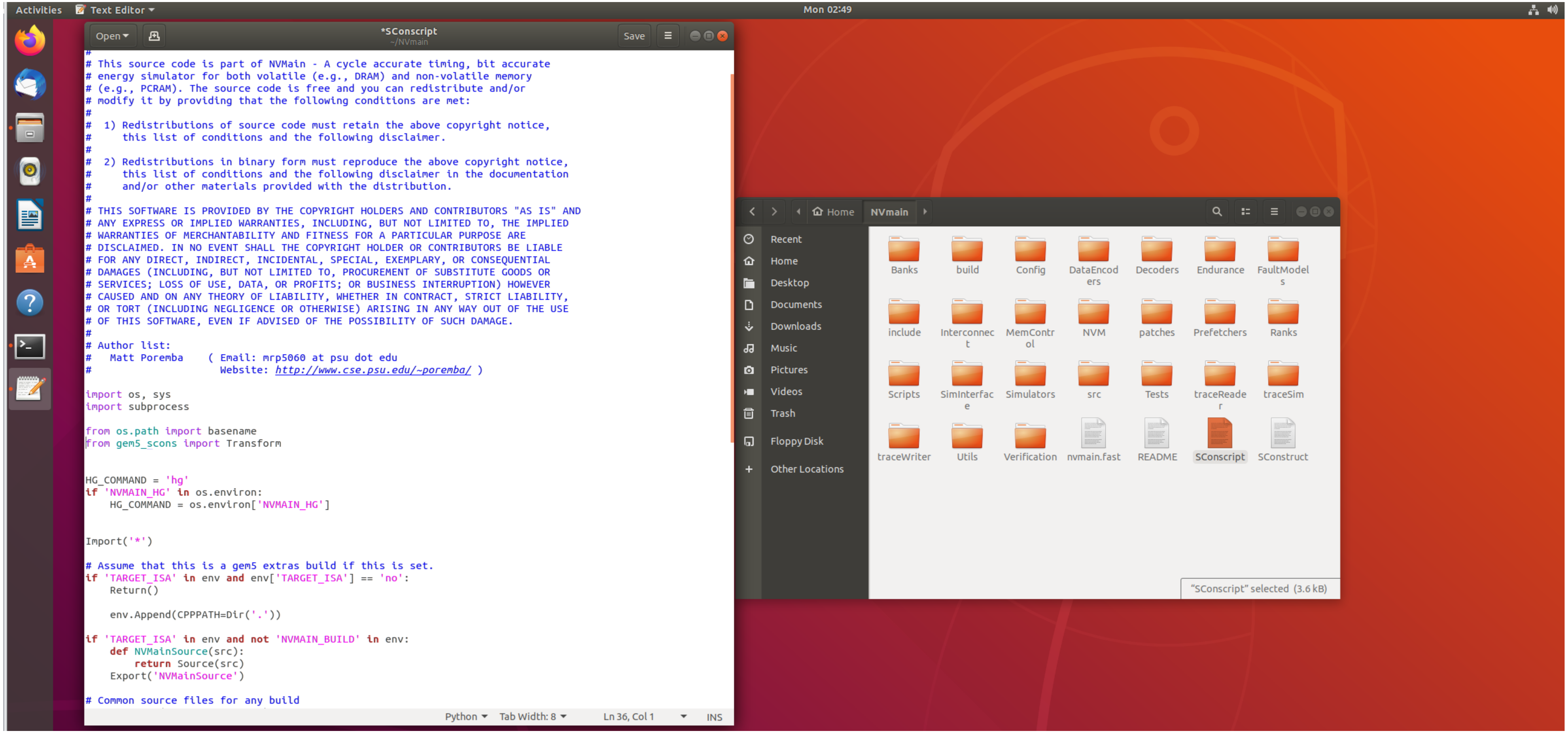
MODIFY GEM5 OPTIONS

- In gem5/configs/common/Options.py Add the next program in line 133 (remember to save it)
- for arg in sys.argv:
-   if arg[:9] == "--nvmain-":
-           parser.add_option(arg, type="string", default="NULL", help="Set NVMain configuration value for a parameter")



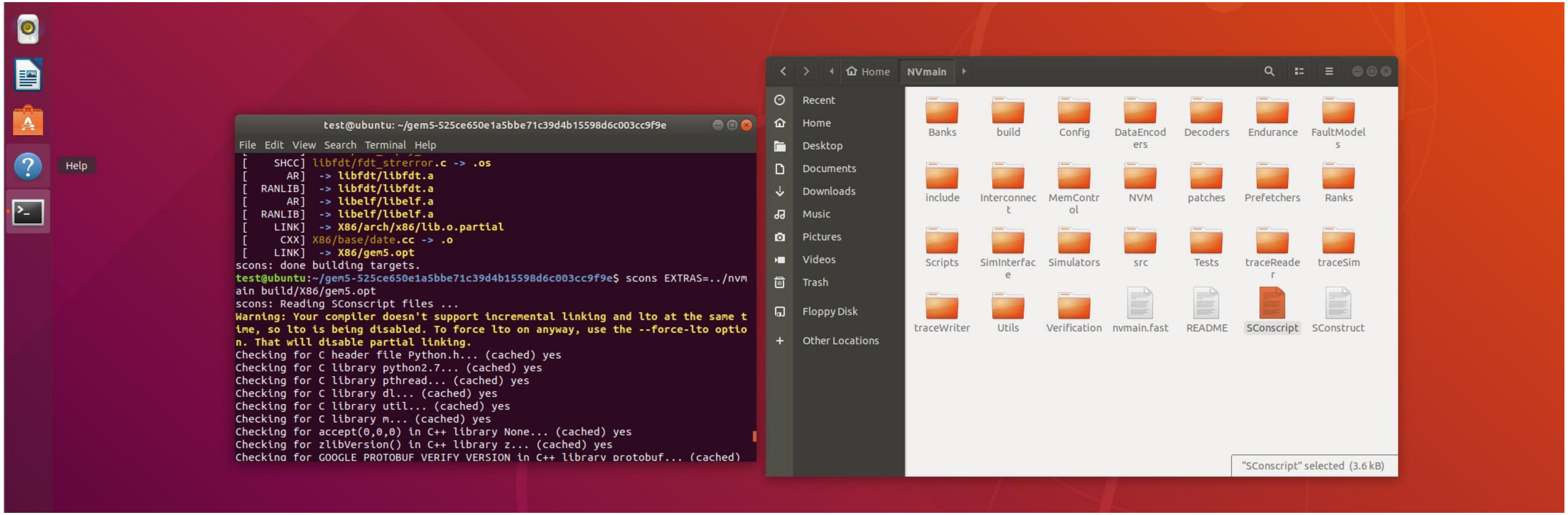
RECOVER SCONSCRIPT

- Restore the previous instructions annotated by nvmain sconscript



COMPILE GEM5

- Compile GEM5 with mixed NVMAIN in the GEM5 directory
- scons EXTRAS=../NVmain build/X86/gem5.opt



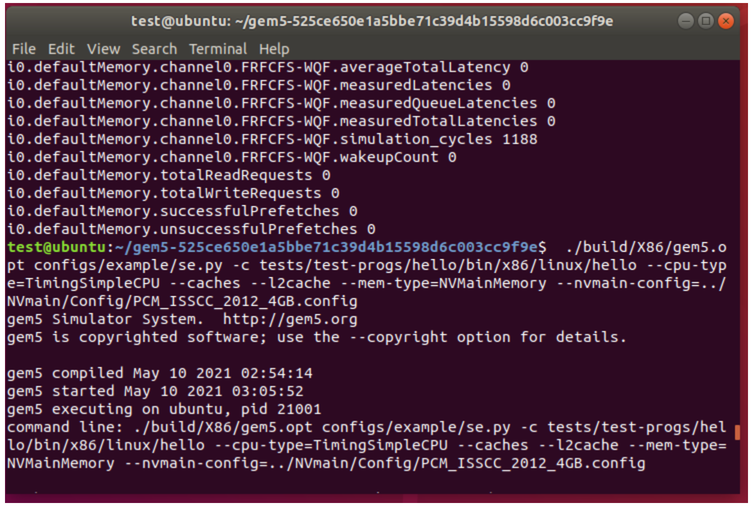
HELLO WORLD

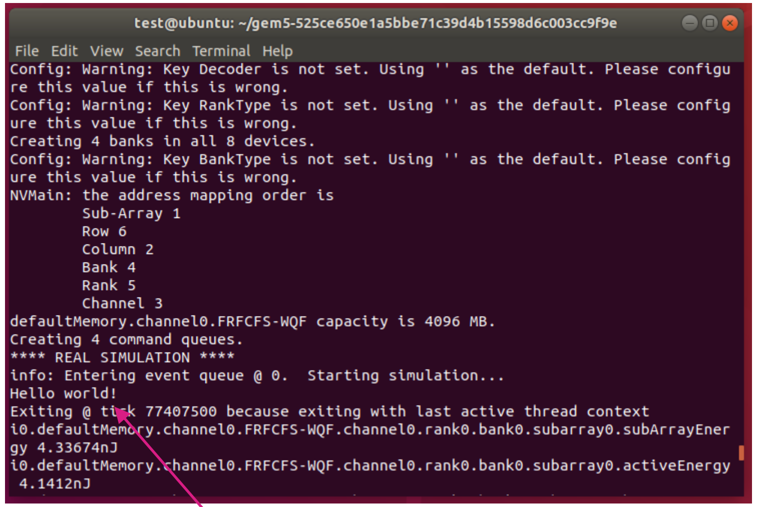
Gem5

code

- ./build/X86/gem5.opt configs/example/se.py -c tests/test-progs/hello/bin/x86/linux/hello --cpu-type=TimingSimpleCPU --caches --l2cache --mem-type=NVMainMemory --nvmain-config=../NVmain/Config/PCM_ISSCC_2012_4GB.config

nvm

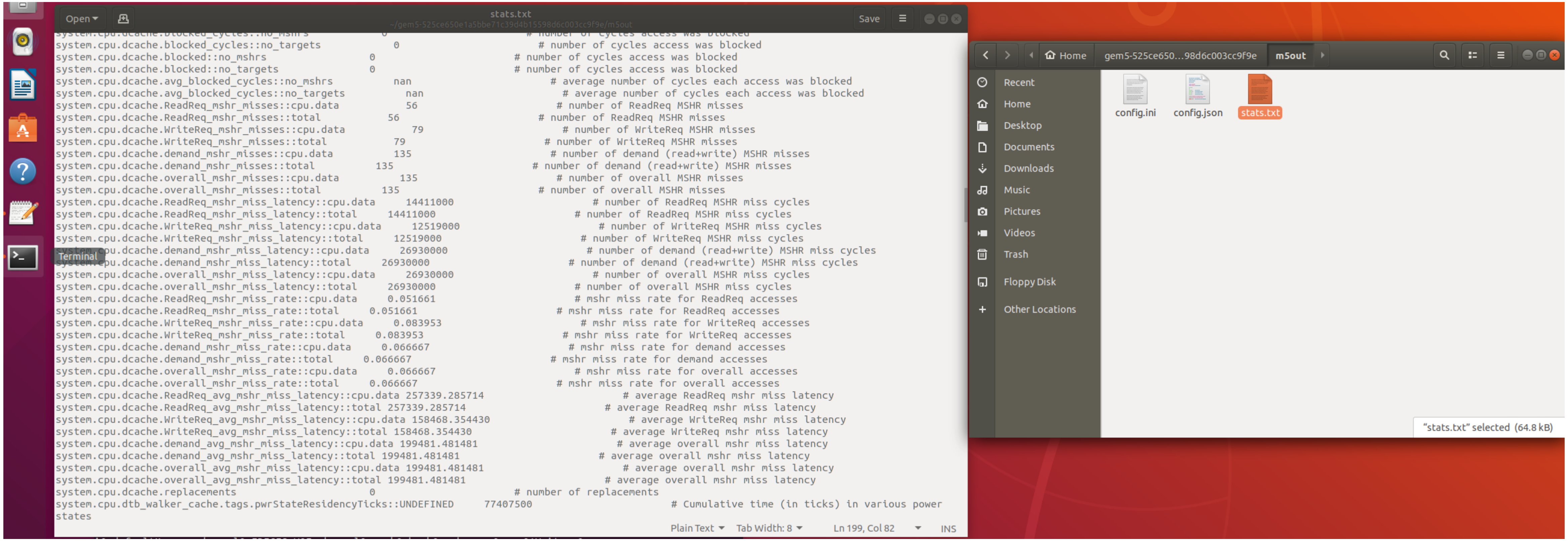




Hello world!

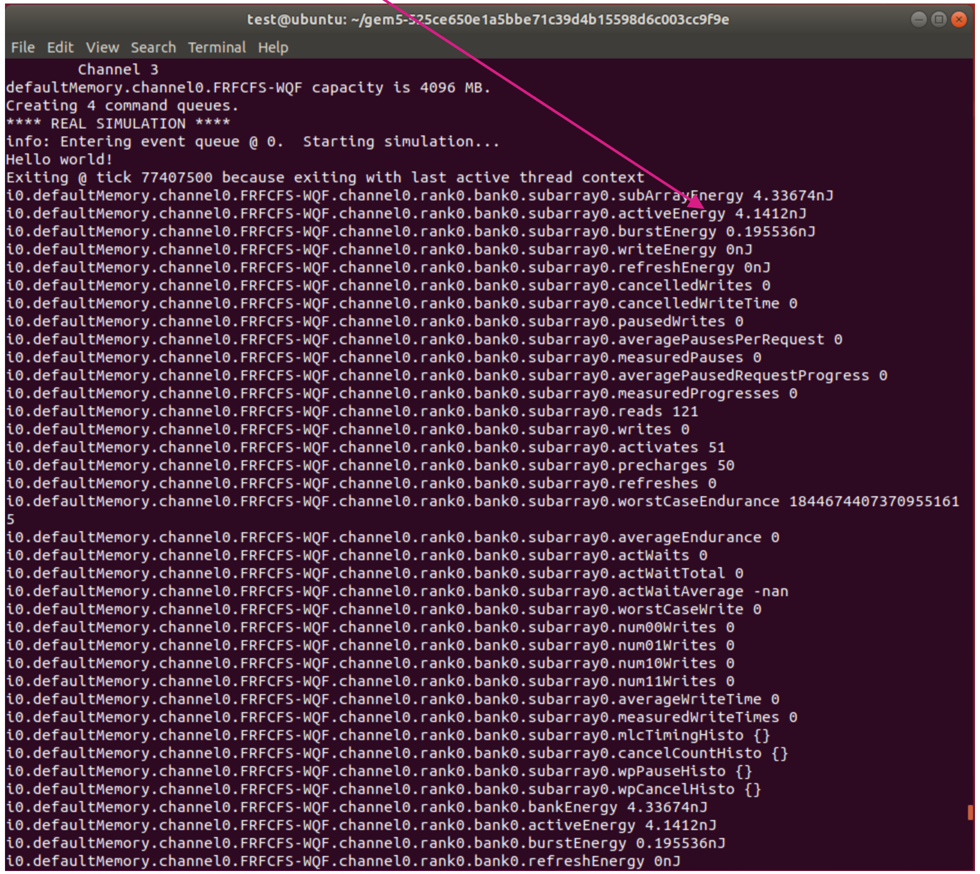
OUTPUT

- You can check the number of cache hits from stat.txt in gem5/m5out



ACTIVE ENERGY

- After running the program, you can see it in terminal.



```
test@ubuntu: ~/gem5-525ce650e1a5bbe71c39d4b15598d6c003cc9f9e
File Edit View Search Terminal Help
        Channel 3
defaultMemory.channel0.FRFCFS-WQF capacity is 4096 MB.
Creating 4 command queues.
**** REAL SIMULATION ****
info: Entering event queue @ 0.  Starting simulation...
Hello world!
Exiting @ tick 77407500 because exiting with last active thread context
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.subArrayEnergy 4.33674nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.activeEnergy 4.1412nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.burstEnergy 0.195536nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.writeEnergy 0nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.refreshEnergy 0nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.cancelledWrites 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.cancelledWriteTime 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.pausedWrites 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.averagePausesPerRequest 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.measuredPauses 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.averagePausedRequestProgress 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.measuredProgresses 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.reads 121
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.activates 51
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.precharges 50
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.refreshes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.worstCaseEndurance 1844674407370955161
5
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.averageEndurance 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.actWaits 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.actWaitTotal 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.actWaitAverage -nan
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.worstCaseWrite 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.num00Writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.num01Writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.num10Writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.num11Writes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.averageWriteTime 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.measuredWriteTimes 0
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.mlcTimingHisto {}
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.cancelCountHisto {}
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.wpPauseHisto {}
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.subarray0.wpCancelHisto {}
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.bankEnergy 4.33674nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.activeEnergy 4.1412nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.burstEnergy 0.195536nJ
i0.defaultMemory.channel0.FRFCFS-WQF.channel0.rank0.bank0.refreshEnergy 0nJ
```

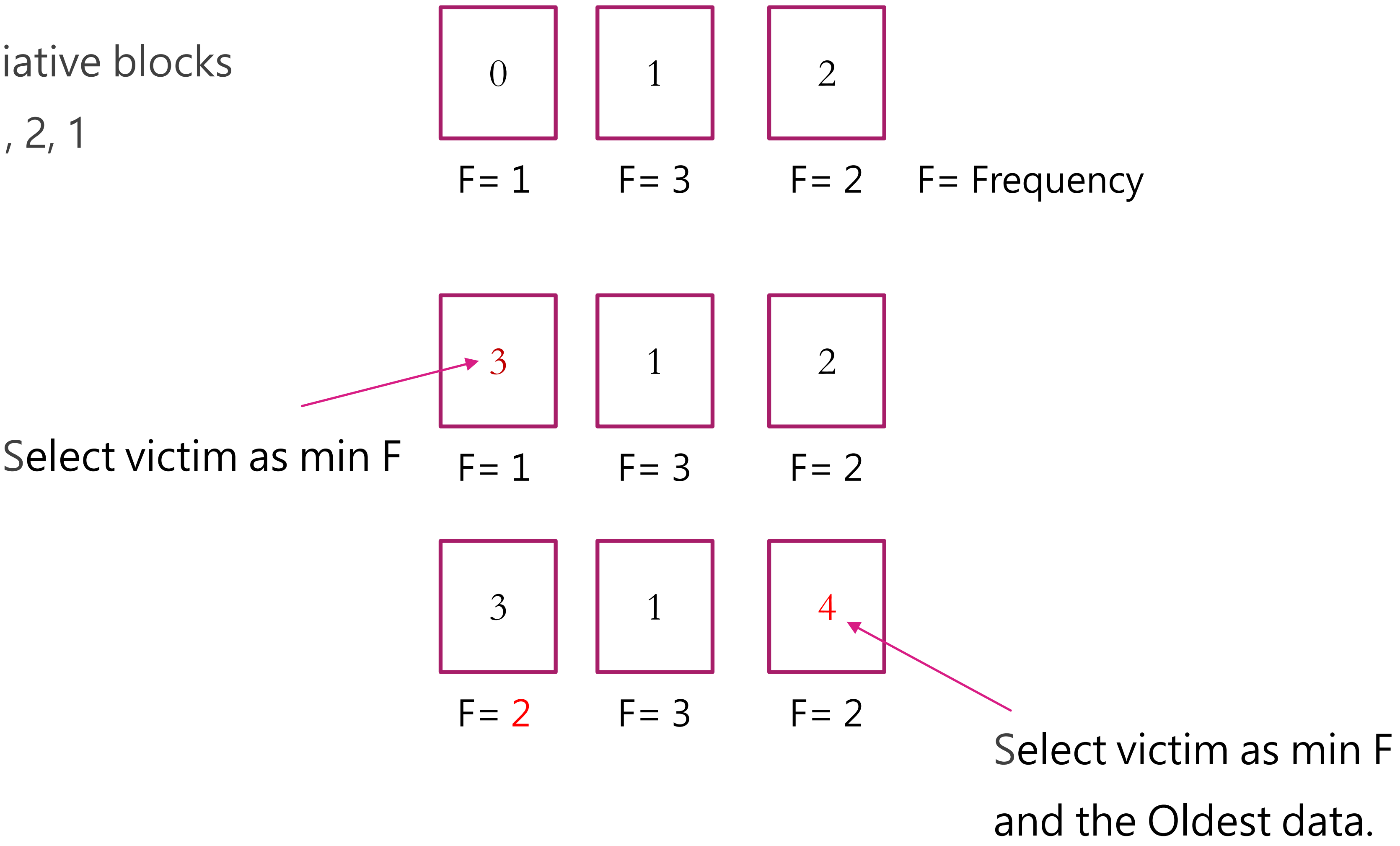
ENABLE L3 LAST LEVEL CACHE

- The files that need to be modified are in the gem5 folder
- Options.py
- Caches.py
- Xbar.py
- BaseCPU.py
- CacheConfig.py
- The first four files only increase the parameters of the L3 cache, and you can imitate them according to the settings of the L2 cache.
- CacheConfig.py needs to connect the L3 cache to the entire Gem5 system. Pay attention to the relationship between the L2 and L3 caches. The system can only use the L3 cache when the L2 cache is already used. This condition is not met.

- For the details of the code, students can go online to find resource keywords like Gem5 L3 cache.

FREQUENCY BASED REPLACEMENT POLICY

- Cache size = 3 full associative blocks
- block request 0 , 1 , 2 , 1, 2, 1
- block request 3
- block request 3, 4



WRITE BACK AND WRITE THROUGH

- You need to understand write back and write through mechanisms.

- You can go to the mem/cache folder to find two files, one is called cache.cc and the other is called base.cc. You can study how the underlying code of gem5 writes data into Memory.